Project Title	**Supply Chain**
Tools	Power BI, MS Excel, DAX
Technologies	Business Analyst
Project Difficulties Level	Intermediate

Dataset: The dataset is available in the given link.

Dataset Link

Project Overview

The goal of this project is to analyze global supply chain metrics across various product types and logistics channels using Power BI to provide stakeholders with actionable insights into operational efficiency and cost management.

About Dataset Content

The dataset contains key information regarding the lifecycle of products, from manufacturing to delivery. Key columns included are:

- **Product Type:** The category of goods sold (e.g., Skincare, Haircare, Cosmetics).
- **SKU:** Unique identifiers for specific stock units.
- **Revenue Generated:** The total sales value produced by each SKU.
- **Number of Products Sold:** The total quantity of units moved.
- **Manufacturing Costs:** The direct cost required to produce the goods.
- **Lead Times:** The duration between ordering and receiving products.
- **Shipping Costs & Carriers:** The expenses and companies involved in transporting goods.
- **Defect Rates:** The percentage of products that failed quality inspections.

Power BI Code

KPI Cards

1. Total Revenue

- **Logic:** This replaces "Total Gross Sales" from your finance report to show the total money coming in.

- **DAX Measure:** Total Revenue = SUM('supply_chain_data'[Revenue generated])
- **Visual Setup:** Drag this into a Card visual. Based on your CSV, this will show the health of your sales.

2. Total Manufacturing Cost

- **Logic:** This acts like the "COGS" (Cost of Goods Sold) in your finance report. It tracks how much you are spending to make the products.
- **DAX Measure:** Total Cost = SUM('supply_chain_data'[Manufacturing costs])
- **Visual Setup:** Drag this into a second **Card** visual.

3. Average Defect Rate

- **Logic:** Instead of "Total Profit," supply chain managers look at quality. High defect rates mean big problems.
- **DAX Measure:** Avg Defect Rate = AVERAGE('supply_chain_data'[Defect rates])
- **Visual Setup:** Use a **Card** visual and format the value as a **Percentage (%)** in the formatting pane.

4. Total Products Sold

- **Logic:** This is the supply chain version of "Total Units Sold" from your internship doc.
- **DAX Measure:** Total Items Sold = SUM('supply_chain_data'[Number of products sold])
- **Visual Setup:** Drag this into your fourth **Card** visual.

Key Charts

A. Map (Geographic Distribution)

This replaces the "Sales by Country" bar chart with a more interactive geographic view.

- **Visual Type:** Map
- **Location:** Drag the Location column (cities like Mumbai, Delhi, etc.) here.
- **Bubble Size:** Drag the **Total Revenue** measure here.
- **Purpose:** Quickly see which cities are generating the most revenue.

B. Donut Chart (Logistics Breakdown)

Use this to see how your shipping budget is distributed among different carriers.

- **Visual Type:** Donut Chart
- **Legend:** Drag Shipping carriers (Carrier A, B, C) here.
- **Values:** Drag the **Total Shipping Cost** measure here.

- **Purpose:** Identify which carrier is the most expensive or handles the most volume.

C. Clustered Bar Chart (Cost vs. Revenue)

This provides a direct comparison of your production costs versus the income earned per product type.

- **Visual Type:** Clustered Bar Chart
- **Y-Axis:** Drag Product type (Haircare, Skincare, Cosmetics) here.
- **X-Axis:** Drag both **Total Revenue** and **Total Manufacturing Cost** here.
- **Purpose:** Visualize the profit margin for each product category.

D. Line Chart (Lead Time Variability)

Since your dataset doesn't have dates, use this to see how lead times fluctuate across different products.

- **Visual Type:** Line Chart
- **X-Axis:** Drag the SKU column here.
- **Y-Axis:** Drag the **Avg Lead Time** measure here.
- **Purpose:** Identify specific products (SKUs) that have unusually long lead times, indicating potential supply chain bottlenecks.

Slicers

Location (Geographic Filter)

This is the most important slicer for your **Map visual**. It allows you to see supply chain performance by specific cities.

- **Column to use:** Location
- **Data values in your file:** * Mumbai
    - Kolkata
    - Delhi
    - Bangalore
    - Chennai

Product Category (Inventory Filter)

This works like the "Product" slicer in your finance report. It helps you see which category is causing delays or high costs.

- **Column to use:** Product type
- **Data values in your file:** * Haircare
    - Skincare
    - Cosmetics

Shipping Carrier (Logistics Filter)

This is essential for your **Donut Chart**. It helps you evaluate which company is handling your transport.

- **Column to use:** Shipping carriers
- **Data values in your file:** * Carrier A
    - Carrier B
    - Carrier C

Dashboard Creation

Step 1: Data Preparation & Loading

- **Importing:** Open Power BI Desktop, click Get Data > Text/CSV, and select supply_chain_data.csv.
- **Cleaning:** Click Transform Data to ensure Revenue, Costs, and Defect rates are set to Decimal Number format, and Location is recognized as a Text category.

Step 2: Building the "Top Layer" (KPI Cards)

We created 4 measures using DAX to show high-level performance:

1. **Total Revenue:** SUM(supply_chain_data[Revenue generated])
2. **Total Items Sold:** SUM(supply_chain_data[Number of products sold])
3. **Avg Defect Rate:** AVERAGE(supply_chain_data[Defect rates])
4. **Manufacturing Cost:** SUM(supply_chain_data[Manufacturing costs])

Step 3: Building the "Middle Layer" (Analytical Charts)

We used four different visuals to provide a 360-degree view of the supply chain:

1. **Map (Global Footprint):**
    - **Visual:** Map
    - **Location:** Location column.
    - **Bubble Size:** Revenue generated.
2. **Donut Chart (Logistics Cost):**
    - **Visual:** Donut Chart
    - **Legend:** Shipping carriers.
    - **Values:** Shipping costs.
3. **Clustered Bar Chart (Product Profitability):**

    - **Visual:** Clustered Bar Chart
    - **Y-Axis:** Product type.
    - **X-Axis:** Revenue generated and Manufacturing costs.
4. **Line Chart (Efficiency Trends):**
    - **Visual:** Line Chart
    - **X-Axis:** SKU.
    - **Y-Axis:** Lead times. (This shows which specific products take the longest to move).

Step 4: Adding Interactive Filters (Slicers)

To make the report interactive like your finance project:

- **Action:** Click the Slicer icon three times.
- **Fields:** Add Location, Product type, and Shipping carriers to individual slicers.
- **Formatting:** In the Format Pane, change the style to Dropdown. Align these at the bottom of the page.

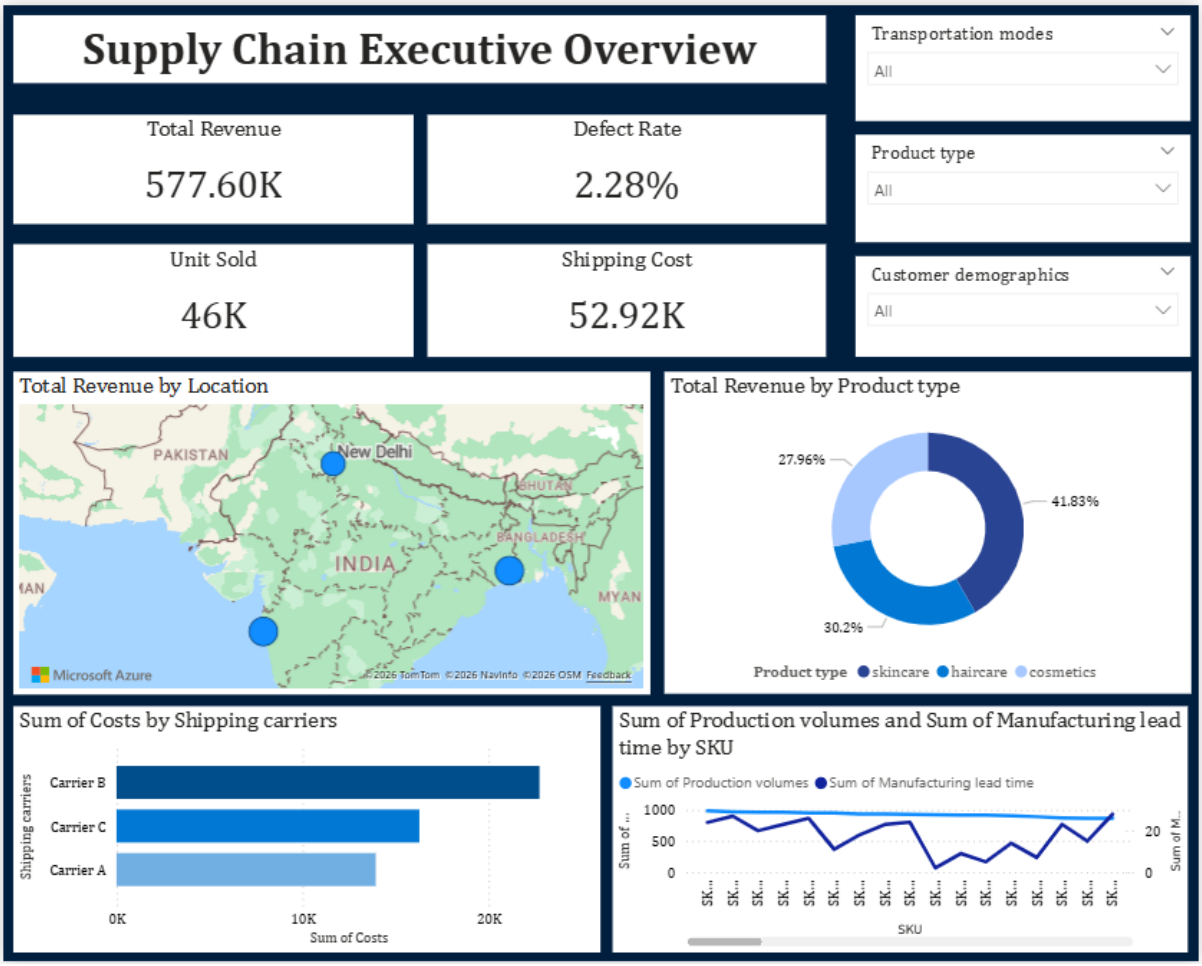
Step 5: Formatting & Design

- **Theme:** Used a clean, professional background with a bold "Supply Chain Analysis" title at the top.
- **Icons:** Added small icons (trucks, factories, coins) next to the KPI cards to improve the UI/UX.

Conclusions

This dashboard transforms raw logistics data into a strategic tool. It allows stakeholders to instantly see which Product Types are most profitable, which Suppliers have the longest lead times, and which Transportation Modes are causing the highest defect rates. By using the interactive slicers, management can filter down to specific cities (like Mumbai or Delhi) to solve local supply chain bottlenecks.

Output


Supply Chain Executive Overview
Transportation modes
All
Total Revenue
577.60K
Defect Rate
2.28%
Product type
All
Unit Sold
46K
Shipping Cost
52.92K
Customer demographics
All
Total Revenue by Location
PAKISTAN
New Delhi
BHUTAN
BANGLADESH
INDIA
MYAN
Microsoft Azure
Total Revenue by Product type
27.96%
41.83%
30.2%
Product type skincare haircare cosmetics
Sum of Costs by Shipping carriers
Shipping carriers
Carrier B
Carrier C
Carrier A
0K
10K
20K
Sum of Costs
Sum of Production volumes and Sum of Manufacturing lead time by SKU
Sum of Production volumes
Sum of Manufacturing lead time
1000
500
0
20
0
SKU